Capital Compensation: A Data-Driven Analysis of Public Salary in Washington, D.C.

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01 Objective

This study aims at investigating and analyzing the public salaries data available in Washington, D.C. The objective of this research is to determine the key factors that influence salaries such as the Grade and the Appointment Type and to make comparisons of salaries within large educational institutions, such as DCPS, and specialized institutions dealing with law or finances.

02 Introduction

This study makes use of data that is available from the Open Data Portal of Washington D.C. and is based on publicly available salaries. It consists of more than 11,000 entries of individuals working at different government agencies with variables such as Agency Name, Job Title, Grade, Salary (Comprate), and Start Date. This data provides important insights into how much public money is spent on human resources in expensive regions like D.C.

03 Data Understanding and Cleaning

The raw data required several preparation steps:

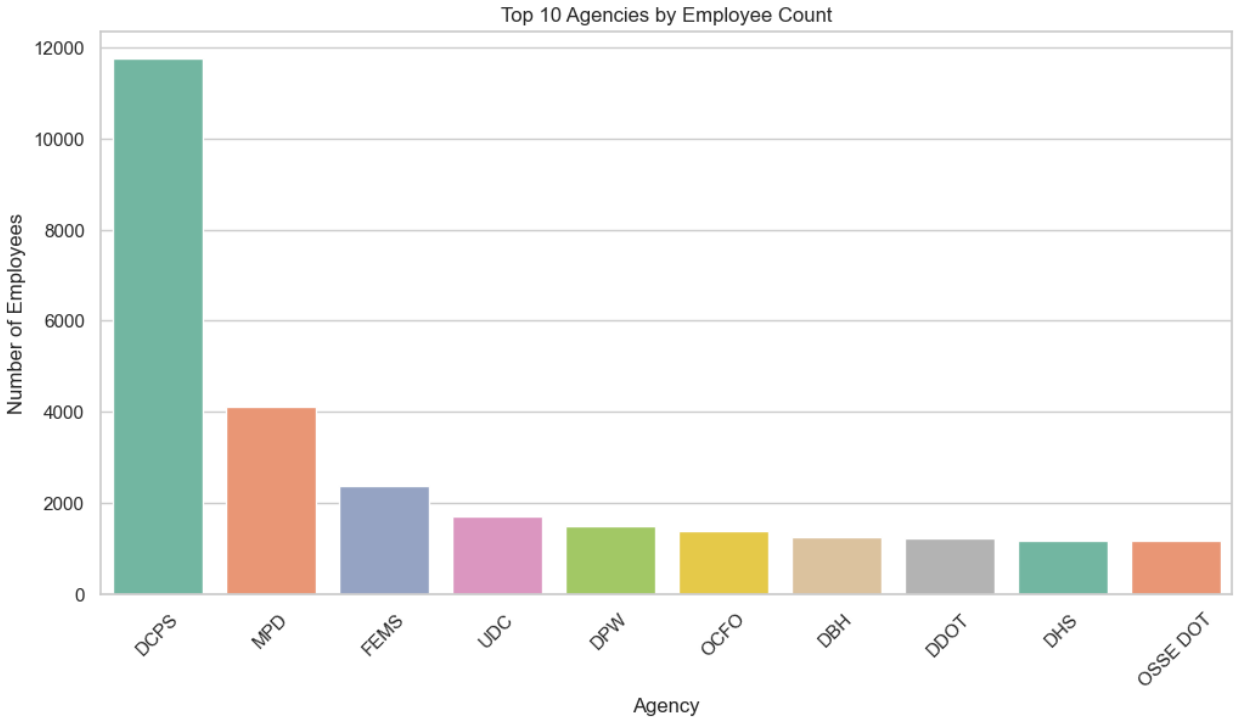
```
df['grade_numeric'] = df['grade'].astype(str).str.extract('(\d+)').astype(float)

plt.figure(figsize=(10,6))
sns.scatterplot(
    data=df,
    x= 'grade_numeric',
    y='comprate',
    alpha=0.4,
    color="#DD8452"
)
plt.title("Salary vs Grade")
plt.xlabel("Grade (Numeric)")
plt.ylabel("Salary")
plt.show()
```

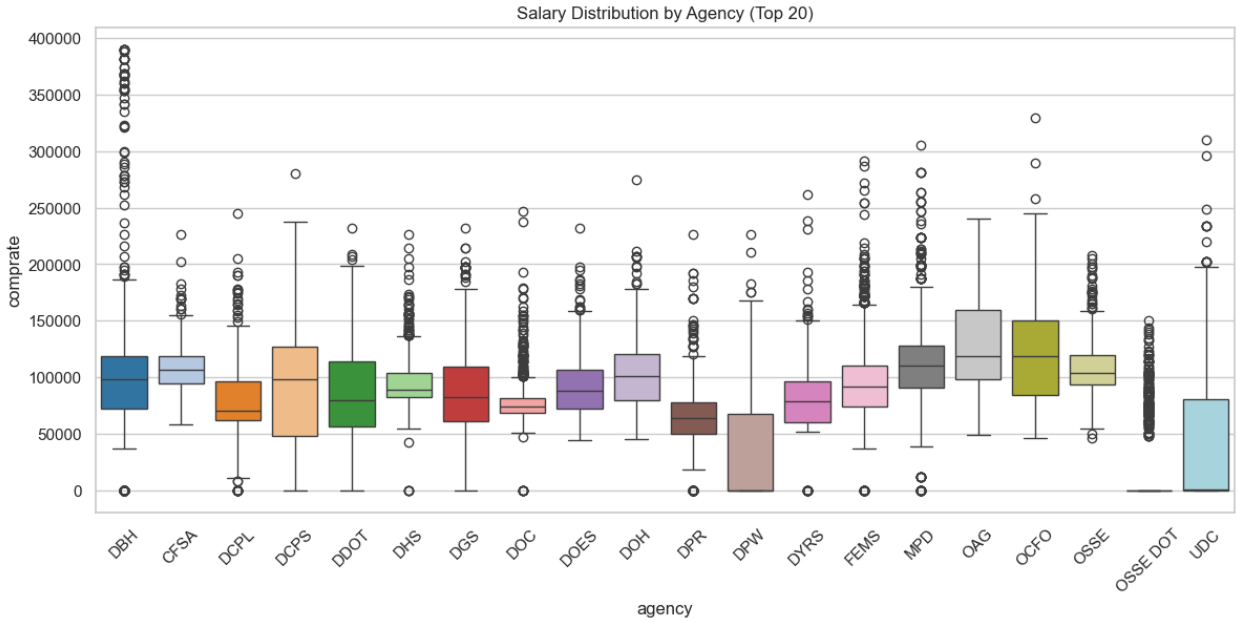
- Feature Engineering: A 'grade_numeric' column was created to convert categorical grade levels into a format suitable for statistical correlation.
- Type Conversion: Salary data was cleaned and ensured to be in a numeric format for calculation.
- Data Integrity: The cleaning process focused on handling missing values and standardizing agency names to allow for accurate grouping and comparison.

04 Storytelling with Visualizations

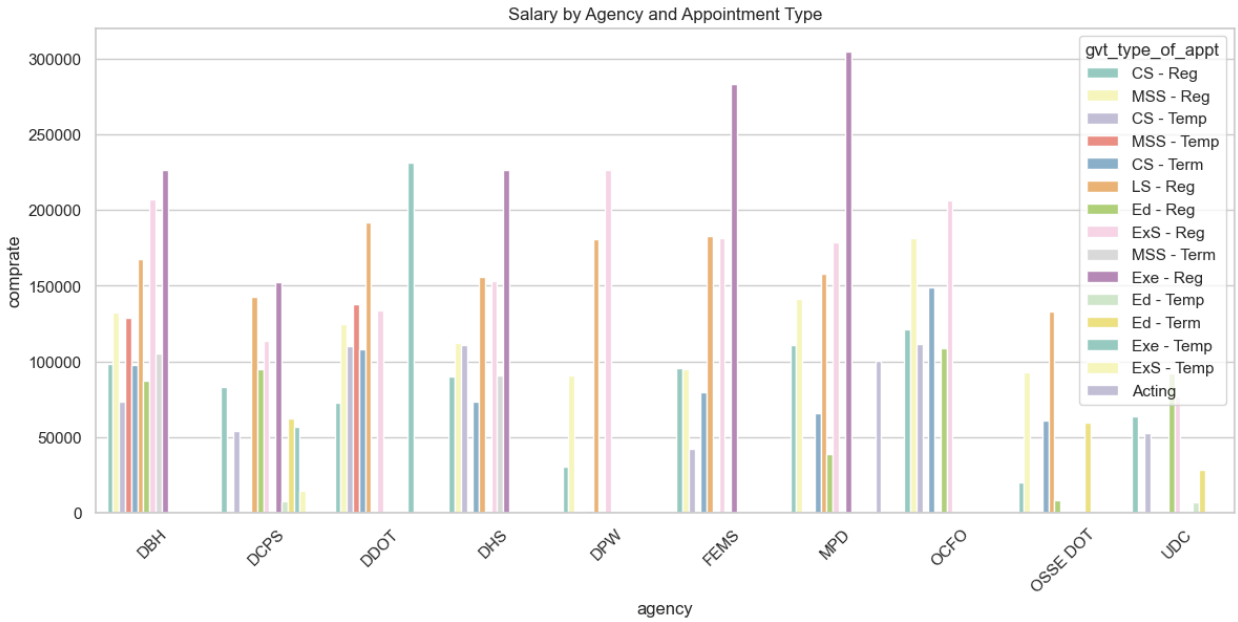
Analysis of the cleaned data revealed several distinct patterns in D.C. government compensation:



Agency Scale vs. Individual Salary: DCPS (District of Columbia Public Schools) employs the largest number of people, contributing a large percentage to the labor force. DCPS has a very large size in terms of employees, but its salaries differ widely.



- High-Value Sectors: The OCFO, OAG, and MPD, for example, pay a much higher salary than what is offered by the DCPS. This is because unlike the widely distributed employees of DCPS, these agencies employ fewer individuals, who thus earn relatively high salaries.



- Salary Drivers: There is a high correlation between the Grade level of an individual employee and their compensation. Data visualizations show that Grade level and certain Appointment Types can be counted upon to accurately predict salaries.

05 Conclusion and Quality of Insights

The wage structure of the general population of D.C. is a compromise between the vast number of teachers and the lucrative specialized industry.

Key Findings:

- Higher grades tend to have higher wages.
- Appointment type plays a crucial role in determining average wages.

- Trends in hiring new workers indicate differences in wage structures from those of older employees.

06 Limitations and Future Directions

Data Limitations: There is no demographic data (race and gender), hence there cannot be any equity or representation study conducted. Also, some job titles are not standardized, thus causing noise in the categorization process.

Suggestions for Further Research: In the future, this research can be expanded by conducting prediction modeling to predict salaries from job features. Clustering of jobs from various agencies can be done to find common patterns among jobs in the public sector. Comparing salaries from the private sector for comparable positions will reveal whether D.C. offers competitive salaries to attract top talent.

07 References

- Washington DC Open Data Portal: Public Salary Records.
- MATH 014 Course Materials and Guidelines.

08 Acknowledgments

Thank you Professor for all the knowledge that was taught this semester about Data Science!